

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5

Total Marks:50

Annexure A SENARIO BASED QUESTIONS

Criteria	Understandin g of Scenari (2)	Identification of Key Issues (2)	Applicatio n of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	Total(10)
Weightag e	20%	20%	25%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I KONGARA JOGESH(192525035)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.JOGESH

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics

Step 1: Identify Data Quality Problems

The dataset contains missing values in Age, Monthly_Spend and Purchase_Frequency. It also contains inconsistent Gender values (M, Male, male, FEMALE, F) and outliers such as Age=105 and Monthly_Spend=500.

Step 2: Handle Missing Values

Age: Missing at C2.

Monthly_Spend: Missing at C3.

Purchase_Frequency: Missing at C5.

Method 1 – Median Imputation

Arrange values in ascending order and find the median.

Replace Age with 36, Monthly_Spend with 36.5 and Purchase_Frequency with 10.

Reason: Median is not affected by extreme values.

Method 2 – k-NN Imputation

Calculate similarity between customers using available attributes.

Select nearest neighbours.

Replace missing values using the average of neighbouring values.

Reason: Preserves data relationships.

Step 3: Detect Outliers using Z-score

Formula: $Z = (X - \text{Mean}) / \text{Standard Deviation}$.

Any value with $|Z| > 3$ is treated as an outlier.

Age=105 and Spend=500 are outliers.

Step 4: Treat Outliers

Method 1: Capping.

Method 2: Log transformation.

Method 3: Remove only if incorrect.

Preferred: Capping because useful records are retained.

Step 5: Standardize Categorical Data

M, Male, male → Male.

F, Female, FEMALE → Female.

Conclusion:

After preprocessing the data becomes accurate, consistent and suitable for recommendation systems.

Q2. Covariance and Data Reduction

Step 1: Calculate Mean Income and Purchase Value.

Mean Income = 80.625.

Mean Purchase Value = 198.75.

Step 2: Compute Covariance

Find deviation of each value from the mean.

Multiply corresponding deviations.

Add all products.

Divide by number of records.

Covariance ≈ 3088.28 .

Interpretation:

Positive covariance means when income increases, purchase value also increases.

Step 3: Feature Selection

Highly related attributes: Income–Purchase Value, Orders–Purchase Value, Avg_Balance–Income.

Selected features: Age, Income, Purchase_Value, Credit_Score and Orders.

Reason: Removes redundant information and reduces computational cost.

Step 4: Standardization

Formula: $Z = (X - \text{Mean}) / \text{Standard Deviation}$.

Apply to every numeric attribute except Cust_ID and Customer_Class.

Benefits:

Reduces scale differences, improves model accuracy, speeds up machine learning and avoids feature dominance.

Q3. Data Transformation and Discretization

Step 1: Equal-Width Binning

Study Hours:

Minimum=2, Maximum=11.

Bin Width= $(11-2)/3=3$.

Bin1:2–5, Bin2:5–8, Bin3:8–11.

Test Score:

Minimum=40, Maximum=100.

Bin Width= $(100-40)/3=20$.

Bins:40–60, 60–80, 80–100.

Step 2: Equal-Frequency Binning

Arrange values.

Divide into three groups with nearly equal records.

Bin1:2,3,4,5.

Bin2:6,7,8.

Bin3:9,10,11.

Step 3: Concept Hierarchy

Study Hours: Low, Medium, High.

Test Score: Poor, Average, Excellent.

Advantages:

Improves interpretation, reduces noise, simplifies classification and supports decision making.

Q4. Normalization

Data: 200,300,400,600,1000.

Step 1: Min-Max Normalization

Formula= $(X-\text{Min})/(\text{Max}-\text{Min})$.

Min=200, Max=1000.

Results:

200=0.000

300=0.125

400=0.250

600=0.500

1000=1.000

Step 2: Z-score Normalization

Mean=500.

Standard Deviation $\approx$ 282.84.

Formula=(X-Mean)/SD.

Results:

200=-1.06

300=-0.71

400=-0.35

600=0.35

1000=1.77

Conclusion:

Min-Max scales values between 0 and 1.

Z-score measures how far each value is from the mean and is useful for outlier detection.

Q5. Online Shopping Behaviour Analysis

Step 1: Equal-Frequency Binning

Arrange durations:

15,30,45,60 | 120,300,600 | 900,1800,3600.

Each bin contains approximately equal records.

Analysis:

Bin1 represents short visits.

Bin2 represents medium visits.

Bin3 represents long visits.

Step 2: Z-score Normalization

Find mean and standard deviation.

Apply $Z=(X-\text{Mean})/\text{SD}$.

Large values such as 1800 and 3600 obtain high positive Z-scores and are easily identified as extreme observations.

Step 3: Min-Max Normalization

Formula= $(X - \text{Min}) / (\text{Max} - \text{Min})$.

Min=15, Max=3600.

Results:

15=0.000

30=0.004

45=0.008

60=0.013

120=0.029

300=0.080

600=0.163

900=0.247

1800=0.498

3600=1.000

Comparison:

Equal-frequency binning groups customers into balanced categories.

Z-score detects outliers.

Min-Max converts values to a common 0–1 scale for clustering and machine learning.

